

Guessing the active set: a rank obstruction on general convex quadratic programs

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Abstract

Primal–dual active-set methods and block principal pivoting solve a strictly convex quadratic program by guessing a whole active set at once and repairing it from the signs the working-set subproblem returns. For bound constraints the guess is safe: the system solved on a candidate set is a principal submatrix of G^{-1} , positive definite whatever the guess, and that P -matrix property is the hypothesis under which the exchange terminates finitely. For general constraints $C^\top x \geq b$ the hypothesis fails. The system is $C_\mathcal{A}^\top G^{-1} C_\mathcal{A}$, positive definite exactly when $C_\mathcal{A}$ has full column rank, which is a property of the guess and not of the data.

That the plain exchange needs repairing on general constraints is known: Curtis, Han and Robinson recover global convergence by carrying an index set auxiliary to the active-set estimate. What we add is the mechanism and its price. The obstruction is structural rather than a missing anti-cycling rule, because no rule for choosing the next set can recover the bound-constrained theorem when the subproblem on the chosen set may be singular; and the singular guess is reached by the method’s own starting rule, not only by an adversarial one. What keeps such a method usable is that strict convexity makes the Karush–Kuhn–Tucker conditions sufficient, so a candidate set is certified rather than trusted.

The exposure is easy to mistake for a technicality, so we measure it on a shipped solver. Over the 600 instances of four constraint families it never occurs, which is why it can be overlooked. Duplicating columns of C , as any programmatically assembled model may, raises it to 93.3%, and the certified fraction on that family falls from 23% at $n = 12$ to 0% at $n = 200$, against 100% everywhere else.

1 Introduction

The strictly convex quadratic program

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top G x - a^\top x \quad \text{subject to} \quad C^\top x \geq b, \tag{1}$$

with $G \in \mathbb{R}^{n \times n}$ symmetric positive definite, $C \in \mathbb{R}^{n \times m}$ and the first m_{eq} of the m constraints understood as equalities, is solved a great many times a day. Mean–variance portfolio selection [11] is exactly (1); so is every step of a sequential quadratic programming method and every horizon of a linear model predictive controller. For dense instances of small to moderate size the method of choice has for forty years been the dual active-set algorithm of Goldfarb and Idnani [7, 8], which reaches the optimal active set one constraint at a time and needs no phase-one problem.

Walking to the active set is not the only option. The alternative is to guess the whole set at once, solve the equality-constrained subproblem it induces, and repair the guess from the signs

that come back. This is the semismooth Newton method of Hintermüller, Ito and Kunisch [9] read as a primal–dual active-set strategy, and block principal pivoting on the associated linear complementarity problem [10, 12, 4] read as a rule for exchanging several indices at a time. Where the walk takes a number of iterations that grows with n , the guess converges in a handful of repairs almost independently of n ; Section 6 measures the gap.

For bound constraints the exchange comes with a theorem. This note is about what happens to that theorem when the bounds are replaced by general linear constraints, and about what an implementation is left with instead.

1.1 What is already known

That the plain block exchange can fail on general convex quadratic programs is established. Curtis, Han and Robinson [5] give a primal–dual active-set framework that is globally convergent on this problem class, obtained by carrying an index set auxiliary to the active-set estimate which holds the indices that would otherwise cycle; the framework exists because plain primal–dual active set does not converge without it. Their repair is the reference point for everything below, and it is the reason the contribution here is a mechanism and not an existence result.

Related bounds run in a different direction. Forsgren, Gill and Wong [6] prove finite termination under nondegeneracy for paired primal and dual active-set methods in a general setting, and Arnström and Axehill [1] compute the exact worst-case iteration count of an active-set method on a parametric family of quadratic programs by enumerating the subproblem sequences it visits. Both concern methods that walk. Neither speaks to a method that guesses.

1.2 What this note adds

1. **The mechanism.** On bound constraints the working-set system is a principal submatrix of G^{-1} , hence positive definite whatever the guess, so G^{-1} is a P -matrix; that is the hypothesis under which block principal pivoting with a least-index guard terminates finitely at the unique solution [10, 12]. On general constraints the system is $C_{\mathcal{A}}^{\top} G^{-1} C_{\mathcal{A}}$, which is positive definite exactly when $C_{\mathcal{A}}$ has full column rank. Rank of a submatrix of C is a property of the *guess*: a guess may name $k > n$ constraints, or $k \leq n$ dependent ones. There is no P -matrix to appeal to, and correspondingly no theorem to inherit (Section 4).
2. **The obstruction is structural.** It is not a missing anti-cycling rule. A rule chooses which set to try next, and no such rule can recover the bound-constrained theorem when the subproblem on the chosen set may be singular. This is consistent with [5], whose repair changes what the method carries rather than the order in which it exchanges, and we read that agreement as evidence for the diagnosis rather than against the claim.
3. **The singular guess is reachable by the method’s own rule.** The standard starting guess is the equality constraints together with whatever the unconstrained minimiser violates. Duplicated columns of C are violated together, so the starting guess names both copies and is dependent before a single repair has been performed (Proposition 4.2). No adversary is required.
4. **What replaces the guarantee.** Strict convexity makes the Karush–Kuhn–Tucker conditions sufficient, so a candidate pair (x, λ) can be checked rather than trusted (Section 5). A method with no convergence theorem is safe when it recognises its own success and falls back when it does not: it returns the minimiser or nothing.

5. **The price, measured.** The exposure is invisible on the random families one would naturally test and unavoidable on a family one would naturally build (Section 6).

We claim no new algorithm and no new convergence theory. The observation in item 1 turns on a rank condition that is already the hypothesis of the standard subproblem solution (Proposition 2.2), and its proof is two lines. Its interest is that it identifies what is lost, says where in the method it is lost, and shows what the loss costs on problems that are not contrived.

1.3 Notation

Vectors are columns. For an index set $\mathcal{A} \subseteq \{1, \dots, m\}$ of size k we write $C_{\mathcal{A}} \in \mathbb{R}^{n \times k}$ for the matrix of the corresponding constraint normals, $b_{\mathcal{A}}$ for the corresponding right-hand sides, and c_i for the i -th column of C . The linear term in (1) is subtracted, so the unconstrained minimiser is $x_u = G^{-1}a$, and constraints are held column-wise, so the program reads $C^\top x \geq b$; both conventions are inherited from the reference implementation of [8] and neither affects anything below beyond signs.

2 The program and the sufficiency of its optimality conditions

Write (1) as

$$\min_{x \in \mathbb{R}^n} f(x) := \frac{1}{2}x^\top Gx - a^\top x \quad \text{subject to} \quad \begin{cases} c_i^\top x = b_i, & i \leq m_{\text{eq}}, \\ c_i^\top x \geq b_i, & i > m_{\text{eq}}, \end{cases} \quad (2)$$

with $G \succ 0$, and let $\Omega = \{x : c_i^\top x = b_i \ (i \leq m_{\text{eq}}), \ c_i^\top x \geq b_i \ (i > m_{\text{eq}})\}$ be the feasible set, a closed convex polyhedron. Since $G \succ 0$ the objective is strictly convex and coercive, so (2) has exactly one solution whenever $\Omega \neq \emptyset$; see [13, Ch. 16].

The Lagrangian is $L(x, \lambda) = \frac{1}{2}x^\top Gx - a^\top x - \lambda^\top (C^\top x - b)$ and the Karush–Kuhn–Tucker conditions are

$$Gx - a = C\lambda, \quad (3)$$

$$c_i^\top x = b_i \quad (i \leq m_{\text{eq}}), \quad c_i^\top x \geq b_i \quad (i > m_{\text{eq}}), \quad (4)$$

$$\lambda_i \geq 0 \quad (i > m_{\text{eq}}), \quad (5)$$

$$\lambda_i (c_i^\top x - b_i) = 0 \quad (i > m_{\text{eq}}). \quad (6)$$

Multipliers of equality constraints are unrestricted in sign. What the argument of Section 5 needs is the converse of these necessary conditions, which we state and prove because every certificate below rests on it.

Proposition 2.1 (Sufficiency). *If (x, λ) satisfies (3)–(6) then x is the unique solution of (2).*

Proof. Let $\tilde{x} \in \Omega$. By strict convexity, $f(\tilde{x}) \geq f(x) + (Gx - a)^\top (\tilde{x} - x)$, with equality only if $\tilde{x} = x$. Substituting (3),

$$(Gx - a)^\top (\tilde{x} - x) = \lambda^\top (C^\top \tilde{x} - C^\top x) = \sum_i \lambda_i [(c_i^\top \tilde{x} - b_i) - (c_i^\top x - b_i)] = \sum_i \lambda_i (c_i^\top \tilde{x} - b_i),$$

the last step by (6) for the inequalities and by $c_i^\top x = b_i$ for the equalities. Each surviving term is non-negative: for $i \leq m_{\text{eq}}$ the bracket vanishes, and for $i > m_{\text{eq}}$ both factors are non-negative by (4) and (5). Hence $f(\tilde{x}) \geq f(x)$ with equality only at $\tilde{x} = x$. $\square$

Proposition 2.1 is what makes an unguaranteed method safe rather than usually right. A point equipped with multipliers passing the four conditions is *the* answer, so a method that produces candidates may be as unprincipled as one likes provided every candidate is tested.

2.1 The working-set subproblem

Fix a working set $\mathcal{A}$ of size k , and write $A = C_{\mathcal{A}}$ for its normals. The *working-set subproblem* holds every constraint in $\mathcal{A}$ as an equality, whatever its type in (2), and ignores the rest:

$$\min_x \frac{1}{2}x^\top Gx - a^\top x \quad \text{subject to} \quad A^\top x = b_{\mathcal{A}}. \quad (7)$$

Proposition 2.2 (Solution of the subproblem). *If A has full column rank k , problem (7) has the unique solution and multipliers*

$$u = (A^\top G^{-1}A)^{-1}(b_{\mathcal{A}} - A^\top x_u), \quad x = x_u + G^{-1}Au, \quad x_u := G^{-1}a. \quad (8)$$

Proof. Stationarity for (7) reads $Gx - a = Au$, so $x = x_u + G^{-1}Au$. Substituting into $A^\top x = b_{\mathcal{A}}$ gives $A^\top x_u + (A^\top G^{-1}A)u = b_{\mathcal{A}}$. The matrix $A^\top G^{-1}A$ is positive definite because $G^{-1} \succ 0$ and A has full column rank, hence invertible, which yields (8); the point so constructed is the unique minimiser by strict convexity of f on the affine feasible set of (7). $\square$

We call $A^\top G^{-1}A$ the *dual Hessian* of the working set. Its invertibility is equivalent to A having full column rank, and that single fact is where the obstruction of Section 4 comes from. Note what the pair (x, u) of (8), extended by zeros off $\mathcal{A}$, does and does not deliver: stationarity (3) and complementarity (6) hold by construction, since the working-set constraints have zero slack and the rest have zero multiplier, while the sign condition (5) on $\mathcal{A}$'s inequalities and primal feasibility (4) off $\mathcal{A}$ need not hold at all. Every method discussed below is a rule for choosing $\mathcal{A}$ so that the two missing conditions come out right.

3 Block exchange on a guessed active set

Given a candidate set $\mathcal{A}$, Proposition 2.2 gives the working-set solution directly:

$$(C_{\mathcal{A}}^\top G^{-1}C_{\mathcal{A}})\lambda_{\mathcal{A}} = b_{\mathcal{A}} - C_{\mathcal{A}}^\top x_u, \quad x = x_u + G^{-1}C_{\mathcal{A}}\lambda_{\mathcal{A}}, \quad x_u = G^{-1}a, \quad (9)$$

with $\lambda_i = 0$ for $i \notin \mathcal{A}$. Both solves go through one Cholesky factorisation of G , computed once, and one of the $k \times k$ dual Hessian, computed per repair. The repair rule reads the two conditions that (9) leaves open, (5) and (4), as instructions:

$$\mathcal{A}' = \{1, \dots, m_{\text{eq}}\} \cup \{i > m_{\text{eq}} : (i \in \mathcal{A} \text{ and } \lambda_i \geq 0) \text{ or } (c_i^\top x < b_i)\}, \quad (10)$$

that is: an active constraint whose multiplier has gone negative does not belong in the set and leaves it; an inactive constraint that is violated at x does belong and joins it; equalities are always in. If $\mathcal{A}' = \mathcal{A}$ then (4)–(6) hold to the tolerance used, and (9) supplies (3) by construction, so the pair is optimal by Proposition 2.1.

The starting guess is the one the rule itself dictates when nothing has been tried:

$$\mathcal{A}_0 = \{1, \dots, m_{\text{eq}}\} \cup \{i > m_{\text{eq}} : c_i^\top x_u < b_i\}, \quad (11)$$

the equalities together with whatever the unconstrained minimiser violates, which is already the answer when it violates nothing.

Two features of this iteration matter for what follows. The whole set is exchanged at once, so the number of repairs is not tied to the number of constraints that have to change status, and in practice it is nearly independent of n (Section 6). And the set is named rather than reached: nothing in (10) or (11) consults the geometry of the normals it selects.

Remark 3.1 (The contrast with the walk). *A dual active-set method modifies $\mathcal{A}$ by one index per step, and the step rules of [8] are exactly what make $C_{\mathcal{A}}$ full column rank for free. A constraint whose normal lies in the span of those already held admits no primal movement, so its step is limited by the multiplier ratio test rather than by the primal one, and it can be added only after enough partial steps have removed the dependence. Linear dependence is therefore unreachable, and no rank test appears anywhere in such an implementation. This is the protection that guessing forfeits, and it is worth naming as a property rather than an accident, because it explains why the difficulty of Section 4 has no counterpart in the classical method.*

4 Where the guarantee comes from, and where it goes

4.1 Bound constraints: the guess cannot be bad

Consider first $\min_{x \geq 0} \frac{1}{2}x^\top Gx - a^\top x$, so that $C = I$ and $m = n$. The system (9) on a candidate set $\mathcal{A}$ is then

$$(I_{\mathcal{A}}^\top G^{-1} I_{\mathcal{A}}) \lambda_{\mathcal{A}} = (G^{-1})_{\mathcal{A}\mathcal{A}} \lambda_{\mathcal{A}} = b_{\mathcal{A}} - x_{u,\mathcal{A}}, \quad (12)$$

whose coefficient matrix is a *principal submatrix* of G^{-1} . Since $G^{-1} \succ 0$, every principal submatrix of it is positive definite, so G^{-1} is a P -matrix and every principal pivot is well defined, whatever the guess. That is the hypothesis under which block principal pivoting with a least-index guard terminates finitely at the unique solution [10, 12, 4], and the guarantee survives inexact inner solves once their residual is small against the problem's decision margin [15]. The guess is safe because no guess can be bad: the set of admissible candidate sets is all 2^n of them.

4.2 General constraints: the guess carries a rank condition

None of that is available for general C . The statement is elementary and we give it in full, because the point is exactly which quantifier moves.

Proposition 4.1 (The dual Hessian of a guess). *Let $\mathcal{A}$ have size k and $A = C_{\mathcal{A}}$. Then $A^\top G^{-1} A \succ 0$ if and only if $\text{rank } A = k$, and $A^\top G^{-1} A$ is singular otherwise.*

Proof. Write $G^{-1} = LL^\top$ with L nonsingular. For $v \in \mathbb{R}^k$, $v^\top A^\top G^{-1} A v = \|L^\top A v\|^2$, which vanishes if and only if $Av = 0$ since $L^\top$ is nonsingular. So $A^\top G^{-1} A \succeq 0$ always, with null space exactly $\ker A$, which is trivial precisely when $\text{rank } A = k$. $\square$

In the bound case A is a set of distinct columns of the identity and $\text{rank } A = k$ holds for every $\mathcal{A}$, which is (12) again. For general C it holds for some $\mathcal{A}$ and fails for others: a guess may name $k > n$ constraints, or $k \leq n$ linearly dependent ones, and then the pivot of (9) is undefined. Rank of a submatrix of C is a property of the guess, not of the data. There is no P -matrix to appeal to, and correspondingly no theorem to inherit.

This is the same rank condition that Proposition 2.2 already carries. What changes between the walk and the guess is not the condition but who discharges it. In the walk it is discharged by the step rules and never tested (Remark 3.1); in the guess it is discharged by nothing and must be tested.

4.3 The bad guess is reached by the method's own rule

One might hope the failure needs an adversary. It does not: the starting guess (11) produces it on constraint sets a modelling language produces by accident.

Proposition 4.2 (A singular guess before the first repair). *Let $p \neq q$ be inequality indices with $c_p = c_q$ and $b_p \leq b_q$. If $c_p^\top x_u < b_p$, then $\{p, q\} \subseteq \mathcal{A}_0$ and $C_{\mathcal{A}_0}$ is rank deficient, so (9) has no solution on $\mathcal{A}_0$.*

Proof. The membership test in (11) puts p into $\mathcal{A}_0$ by hypothesis, and then $c_q^\top x_u = c_p^\top x_u < b_p \leq b_q$ puts q there too. Then $C_{\mathcal{A}_0}$ has two identical columns, so its rank is below its column count, and $C_{\mathcal{A}_0}^\top G^{-1} C_{\mathcal{A}_0}$ is singular by Proposition 4.1. $\square$

Duplication is the most elementary instance and not the only one: what the argument needs is a linear dependence among constraints that the unconstrained minimiser violates simultaneously, so any nontrivial dependence among the normals will do provided x_u violates all of them at once; for normals summing to zero that requires $\sum_i b_i > 0$. Duplicated constraints are not exotic. They arise whenever a model is assembled programmatically from overlapping rules, which is how portfolio and control models are usually built.

Note where the failure sits. It is not that the method converges to the wrong point, or cycles between two sets. The very first subproblem is undefined, before any repair has been performed and therefore before any rule governing repairs has had an opportunity to act.

4.4 Why the obstruction is structural

Block exchange can also cycle, and for cycling the standard remedy is a least-index rule that flips only the lowest-indexed offending constraint, after Bland’s rule for the simplex method and Murty’s least-index rule for linear complementarity problems [12]. In the bound-constrained case that remedy comes with a theorem. The theorem does not survive the generalisation, and the reason is not that the right rule has yet to be found.

A rule of that kind is a map from the history of the iteration to the next candidate set. It can address cycling, which is a property of the sequence of sets. It cannot address singularity, which is a property of a single set: Proposition 4.2 exhibits a failure at the first set the method names, where there is no sequence yet. A rule could in principle avoid singular sets, but only by testing them, which is the factorisation the exchange was meant to make cheap, and only by departing from the set the signs of (10) point at, which is the whole content of the method. Either way what results is no longer the block exchange.

That reading is consistent with the repair the literature offers, and this is the strongest evidence for it we can give. Curtis, Han and Robinson [5] recover global convergence by carrying an index set auxiliary to the active-set estimate, which holds the indices that would otherwise cycle. That changes what the method carries, not the order in which it exchanges. Our claim is that it had to: no reordering recovers the bound-constrained theorem when the subproblem on the chosen set may be singular. We state this as a diagnosis rather than an impossibility theorem, because it quantifies over rules rather than over instances, and a rule that carries extra state is outside its scope by construction.

5 What replaces the guarantee

A method that may fail, and that may fail by stabilising on a set that is simply not optimal, is usable when the answer is checked. Here Proposition 2.1 carries the argument: a candidate (x, λ) satisfying

$$\|Gx - a - C\lambda\|_\infty \leq \varepsilon, \quad |c_i^\top x - b_i| \leq \varepsilon \ (i \leq m_{\text{eq}}), \quad c_i^\top x - b_i \geq -\varepsilon, \quad \lambda_i \geq -\varepsilon, \quad |\lambda_i(c_i^\top x - b_i)| \leq \varepsilon \quad (13)$$

on the inequalities, with ε scaled by the data, *is* the answer to that tolerance. Stationarity is checked against G directly and not inferred from the construction, since the construction is what an ill-conditioned working set corrupts. A candidate that fails is discarded and an exact method runs instead, so guessing can never degrade the answer: it produces the minimiser or nothing.

This is the trade the rest of the note is about. One gives up finite termination, which on general constraints was not available anyway, and buys back correctness by verification. The asymmetry that makes the trade worth taking is that verifying (13) costs $O(nm)$ for a dense C , one pass over the constraints, against the $O(n^3)$ of the fallback solve.

Two implementation points follow, and they are the ones a reader who has to build such a solver will want.

Remark 5.1 (The rank test is the factorisation). *There is no need for a separate rank decision. Attempting the Cholesky factorisation of $C_A^\top G^{-1} C_A$ is the test: by Proposition 4.1 the matrix is positive definite exactly when the guess has full column rank, so a dependent guess fails the factorisation rather than returning something plausible. What an implementation must not do is solve the system by a method that tolerates rank deficiency and returns a least-squares answer, because that converts a detected failure into an undetected one.*

Remark 5.2 (Two tolerances with very different jobs). *The tolerance in (10) decides which set is tried next. A bad choice there costs a repair or a fallback and can never cost correctness, so it is not delicate. The tolerance in (13) is the only thing standing between a non-optimal point and the caller, and is therefore the one to reason about. It is nonetheless not delicate either, because the quantity it thresholds is bimodal: a candidate constructed from a well-conditioned working set satisfies (13) to a few multiples of the machine epsilon, and one built on a set that is not has no reason to come close. Section 6 measures the accepted population, whose worst residual is 2.6×10^{-13} .*

The least-index rule keeps a role under this reading, but a demoted one. It is a way of making progress where block exchange oscillates, and not a termination proof. Neither it nor the rank test substitutes for the guarantee. The certificate does.

6 What the exposure costs

Section 4 says the guarantee is lost. It does not say how often the loss is felt, and that question decides whether the result is a caution or a curiosity. We measure it on a shipped solver, `cvx-quadprog` version 0.4.2 [14], whose fast path is the exchange of Section 3 guarded by the certificate of Section 5, with the classical dual walk as its fallback. The experiment instruments the shipped routines by wrapping them rather than reimplementing them, so what is measured is what runs. It covers 5 constraint families, 30 instances each, at $n \in \{12, 25, 50, 100, 200\}$, with $G = BB^\top + nI$ for B Gaussian and every problem drawn from a seed derived arithmetically from family, size and instance index.

The method works, and works well, where the rank condition holds. On the four well-posed families (bounds, budget-plus-bounds, dense C , and C with equalities) every guess is certified at every size, and the repair count rises only from 1.3 to 3.5 across a sixteen-fold range of n , never exceeding 5. The comparison that matters is against the exact walk on the same instances: on budget-plus-bounds it takes 13 outer iterations at $n = 100$, and its count grows linearly where the repair count is very nearly flat. This is why the exchange is worth having, and it is the reason the negative result matters rather than a qualification of it.

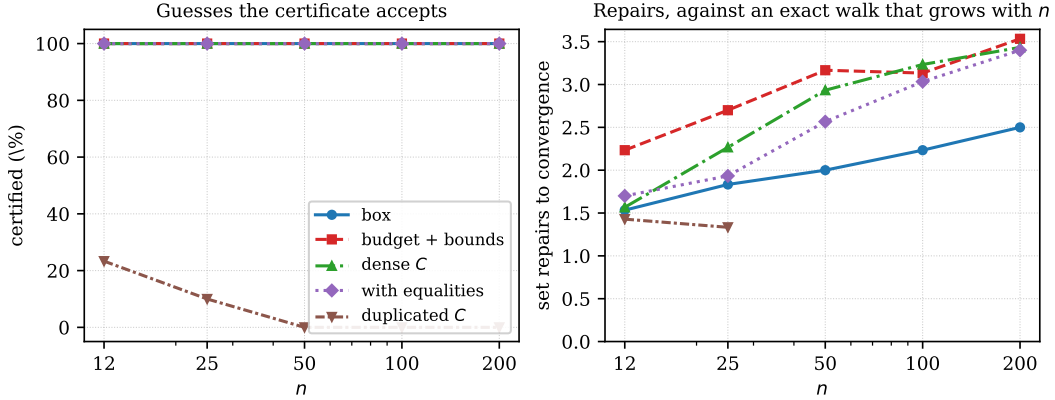


Figure 1: Left: the fraction of guesses the certificate accepts. Right: set repairs to convergence. Four families are certified without exception at every size and settle in 1.3 to 3.5 repairs while the exact walk’s outer iterations grow linearly in n . The fifth is constructed so that a guessed set can be linearly dependent, and it fails.

The rank-deficient guess is real, and invisible by accident. Over the 600 instances of those four families, not one guess produced a rank-deficient working-set system, from which a reader might conclude the exposure is theoretical. It is not; it is invisible in random data, where a dense C almost never has a dependent subset that the guess happens to name. The fifth family makes it reachable in the most elementary way, by repeating columns of C , which is the construction of Proposition 4.2. There the rank test fires on 93.3% of attempts and the certified fraction collapses from 23% at $n = 12$ to 0% at $n = 200$, against 0.0% on the others. The two numbers bracket the practical question: a benchmark suite of random instances will report that the fast path never fails, and a model assembled from overlapping rules will find that it essentially never succeeds.

Where the failures are caught. Of the candidates reaching the certificate, 610 were accepted with a worst KKT residual (13) of 2.6×10^{-13} , and 0 were rejected. We report that asymmetry instead of smoothing it: on these families the certificate never had to reject a converged candidate, because the failures were caught earlier and more cheaply by the Cholesky factorisation refusing a dependent guess. The certificate remains the guarantee, and it is what one must keep, since nothing here bounds how a rank-deficient guess would behave under a solver that tolerated it. On this evidence, though, the rank test does the work in practice.

What this does not establish. The experiments are dense and of modest size, with $G = BB^\top + nI$, so they say nothing about conditioning; the residuals in Remark 5.2 would grow with $\kappa(G)$. That choice also works against the conclusion drawn here rather than for it, since a well-conditioned G spreads the minimiser so that few bounds bind, where a realistic covariance concentrates it and binds many. Nor do they establish that duplication is the common case in practice; they establish that when it occurs the failure rate is not small.

7 Discussion

The result is a small one and it is worth stating its size accurately. Proposition 4.1 is an elementary rank observation with a two-line proof. That the plain block exchange fails on general convex

quadratic programs is known, and repaired, in [5]. What is added here is the location of the failure and the price of ignoring it: the working-set system is positive definite exactly when the guessed normals are independent, which is a property of the guess rather than of the data, so the P -matrix structure that carries the bound-constrained theorem is gone and no reordering brings it back; and the guess that breaks is the one the method's own starting rule produces on duplicated constraints, at a rate of 93.3% where random instances show 0.0%.

Three consequences seem worth drawing.

The bound-constrained intuition does not transfer, and the reason is specific. Block principal pivoting on $x \geq 0$ is safe not because the pivots are small or the sets well chosen but because G^{-1} is a P -matrix, which is a statement about all 2^n candidate sets at once. Reading across from that setting to $C^\top x \geq b$ replaces a statement about the data with a statement about each guess, and the two are not the same kind of hypothesis.

A rank test is not optional, and the factorisation already is one. An implementation that solves (9) by a route tolerant of rank deficiency will return a plausible vector on a dependent guess and has converted a detectable failure into a silent one. Cholesky refuses, which is the behaviour to want.

Certification is the right replacement, not a fallback of last resort. Strict convexity makes the Karush–Kuhn–Tucker conditions sufficient (Proposition 2.1), so the exchange can be run as a heuristic proposer whose output is verified at $O(nm)$ and discarded when it fails. On the families measured, that arrangement gives the iteration count of the guess and the guarantees of the walk, and the two are not in tension because they are sequential rather than alternative. The general form of the observation is that a method which may fail but can recognise its own success needs a certificate and not a convergence theory.

Two limits should be kept in view. The scope here is $G \succ 0$; the dual walk extends to semidefinite G with finite termination by Boland [3] and by proximal regularisation in [2], and we have not asked what the rank condition looks like there, where G^{-1} does not exist and the dual Hessian must be formed differently. And the diagnosis in Section 4.4 quantifies over rules for choosing the next set, not over algorithms: a method that carries state beyond the active-set estimate, which is precisely what [5] does, is outside its scope and is not contradicted by it.

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